

Spatio-Temporal Weighted Graph Reason Learning for Multivariate Time-Series Anomaly Detection

Huaqi Zhang¹, Huaxin Pang¹, Yufeng Zhao¹, Guandong Gao, Yang Song, Shikui Wei²,
and Yao Zhao¹, *Fellow, IEEE*

Abstract—Constructing an efficient and deployable anomaly detection system requires achieving high accuracy, low latency, and reliability. Existing methods either spend considerable time extracting rich spatio-temporal features to enhance anomaly detection performance, or blindly integrate multisource features to boost accuracy, often neglecting the reliability of feature aggregation. The tradeoff between the three objectives must be carefully considered when developing the model. To address these challenges, we introduce a novel Spatio-Temporal Weighted Graph Reasoning Learning (STWGRL) framework for multivariate time-series anomaly detection. Specifically, we propose a series-denoising receptive-weighted key value (D-RWKV) module to efficiently capture and model expressive long-term sequence information through a linear scaling mechanism. D-RWKV ensures compatibility by alleviating the memory bottleneck and enabling parallelized training. Furthermore, we design a targeted-awareness graph adaptive aggregation (TaGAA) module to learn the directed graph and adaptively enhance the signal's intrinsic characteristics. Two graph-constraint losses are employed to strengthen the consistency and sparsity of the learned graphs. Experimental results on multiple benchmark tasks clearly demonstrate the effectiveness of the proposed framework. STWGRL achieves more accurate scores than most baselines, while containing fewer than 10K parameters.

Index Terms—Anomaly detection, graph neural network, graph structure learning, multivariate time series (MTS), temporal modeling.

I. INTRODUCTION

THE Internet of Things (IoT) is a system of interconnected computing devices, mechanical or digital machines,

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Huaqi Zhang and Guandong Gao are with the Department of Information Management, National Police University for Criminal Justice, Baoding 071051, China.

Huaxin Pang and Yufeng Zhao are with the Chinese Medicine Data Center, China Academy of Chinese Medical Sciences, Beijing 100053, China (e-mail: 20112005@bjtu.edu.cn).

Yang Song is with the School of Mechatronics and Electrical Engineering, Hebei Agricultural University, Baoding 071001, China.

Shikui Wei and Yao Zhao are with the School of Computer and Information Technology, Beijing Jiaotong University, Beijing 100044, China.

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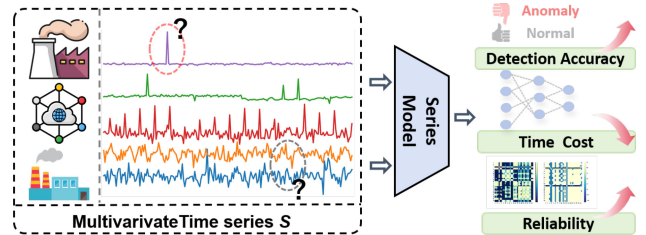


Fig. 1. Schematic diagram of abnormal detection of multiple types of IoT devices. The criteria for successful application of anomaly detection algorithms based on MTS data include: higher detection accuracy, lower time cost, higher reliability.

equipped with unique identifiers and capable of automatically transmitting data over a network. The rapid growth and widespread adoption of IoT systems, particularly in sectors, such as smart factories [1], cloud servers [2], [3], and smart transportation [4]. These IoT infrastructures are becoming increasingly complex and large-scale to meet the demands for intelligence and high efficiency. Consequently, the risks of IoT malfunctions and cyberattacks are increasing dramatically [5]. For instance, a cloud system serving thousands of users is typically vulnerable to various security threats, including tens of thousands of cyberattacks and operational mistakes by technicians. Autonomous factories are also prone to malfunctions due to equipment aging or damage caused by harsh environmental conditions. As a result, numerous sensors are deployed to monitor these systems and detect abnormal activities in real-time.

To detect and address anomalies in a timely manner, monitoring systems collect sensor signals and infrastructure data at extremely short intervals, resulting in vast amounts of multivariate time series (MTS) data. Given the diversity of sensor patterns and the long duration of data recording, it is impractical to rely solely on manual methods and univariate prediction models to detect anomalies. Therefore, engineers have developed numerous algorithms to automatically detect anomalous behavior in MTS. Due to the diversity and extreme sparsity of anomalies in MTS data, current approaches are increasingly adopting unsupervised learning methods [3], [6] to learn the evolutionary representation of the normal sequence from unbalanced datasets. Generally, a well-designed algorithm applicable to IoT systems should achieve the following objectives [7], [8], as illustrated in Fig. 1: achieving higher detection accuracy in MTS with severely imbalanced normal and abnormal states; reducing the time cost associated with detecting anomalies: i.e., achieving lower

temporal overhead); and producing more reliable results that help engineers accurately locate fault positions.

To achieve the above goals in anomaly detection, researchers have developed a series of algorithms and rule-based models to predict anomalies. Classical machine learning methods [9] either use statistical-based rules, such as bias, distance, and density, to estimate the probability of anomalies or employ low-level handcrafted features to train classification models for anomaly detection. Classical methods, such as the Gaussian model and one-class support vector machines, can quickly recognize sequences and identify potential anomalous time steps, but they suffer from low accuracy due to the lack of learned spatio-temporal features. The development of deep learning techniques has made significant progress in information encoding. Deep models can capture more abundant features and sequence changes, integrating this information into higher dimensional representations compared to previous simpler methods. Recurrent Neural Networks (RNNs) and their variant, Long-Short-Term Memory (LSTM) [10], can receive multiple sequence fragments simultaneously and integrate them into a single, dense temporal representation for each signal. The dependencies of long-range time are efficiently learned and embedded into the vector. However, RNNs cannot be parallelized in the time dimension during inference, which restricts their response rate. CrossFuN [11] transferred MTS sequences into time-frequency maps and employed a 1-D convolution layer to extract time features, which requires generating redundant frequency maps and incurs higher time costs. The Transformer framework, prominent in sentence modeling, has also been introduced for temporal representations. With the attention mechanism, the Transformer can capture long-term intrasignal temporal features and intersignal relationships simultaneously. TranAD [12] combined the Transformer with Generative Adversarial Networks (GANs) to detect anomalies through a reconstruction-based model. Autoformer [13] and Anomaly Transformer [7] used attention to model interactions and generate mixed representations. Despite these significant improvements in detection scores, the quadratic complexity inherent in the attention mechanism renders the architecture computationally expensive and memory-intensive for tasks involving long input sequences or prediction-quick situations. Thus, addressing the challenge of modeling key spatio-temporal features while reducing time costs is crucial.

While improving detection speed and accuracy, the reliability of the results has increasingly attracted attention to more accurately locate abnormal events. Existing studies help assess the reliability of results by explicitly modeling the relationships between multiple signals. Meanwhile, prediction accuracy can be enhanced through the propagation and interaction of signal characteristics. These works [14], [15] predefined relevant graphs using several similarity metric functions, neglecting the distinction between self-similarity graphs and task-related graphs. This can degrade the ability of anomaly detection. GDN [16] constructed a static relevant graph using learnable global embeddings of each signal to represent the signals' normal behavior relationships. Additionally, DVGCRN [17] considered the variability in

long-term MTS and the potential change in signal correlations from normal to abnormal timestamps. It designed a deep embedding-guided probabilistic generative network to model hierarchical nondeterministic inter-relationships based on segmented subsequences. DyGraphAD [18] introduced the dynamic time warping (DTW) strategy to extend the receptive field. However, these methods are insufficient for modeling long-range directed dependencies. Moreover, current methods [5] cannot assign more weight to the signal's own features.

To address these challenges, we propose a spatio-temporal weighted graph reasoning learning (STWGRL) framework, named STWGRL, to adaptively and efficiently detect anomalies in MTS data. It achieves a balance between detection precision, time cost, and reliability through its included modules and operations. Specifically, we develop a series-denoising receptance-weighted key value (D-RWKV) module to efficiently encode key temporal features. The series-denoising operations can quickly filter out simple noisy signals. D-RWKV is carefully designed to not only capture features in parallel from long-range sequences, alleviate the memory bottleneck and quadratic scaling associated with Transformers. Additionally, the targeted-awareness graph adaptive aggregation (TaGAA) module is designed to compute the relationships between multiple signals and model the spatial features. Considering the heightened dependence of the signal's intrinsic features, TaGAA designs a novel GNN aggregation operation, i.e., proposing a weighted factor to adaptively preserve the original features. The context consistency constraint is applied during training to ensure the consistency of relationship graphs between neighboring timestamps. Comprehensive experiments demonstrate that our method achieves superior detection performance with a model containing fewer than 10K parameters. To summarize, our main contributions are three-fold.

- 1) We developed a STWGRL framework to efficiently detect complex anomalies in MTS in IoT systems. STWGRL achieves this by modeling temporal dependencies in parallel, while explicitly capturing the inter-relationships between signals.
- 2) A targeted-aware graph adaptive aggregation (TaGAA) module is proposed to reason the dynamic, unbiased related graph and flexibly integrate the signal's characteristics by relaxing information retention into learnable parameters.
- 3) Experiments on six benchmark datasets demonstrate that STWGRL outperforms current techniques. Appended graphs provide a deeper understanding of the signals' behavior and dynamic interactions within a system.

II. RELATED WORK

Over the past decades, significant progress has been made in the development of anomaly detection techniques for IoT systems. Complex IoT environments often involve numerous sensors that monitor changes in intrasystem behavior. As a result, leveraging multiple signals in a comprehensive manner for anomaly detection has proven to be both

efficient and practical. Anomaly detection methods can generally be classified into three categories based on the use of labels during the training process: 1) supervised; 2) semi-supervised; and 3) unsupervised approaches. In this section, we focus on various existing methods for unsupervised MTS anomaly detection, which can be further divided into two primary categories: 1) temporal and 2) spatial feature modeling.

A. Temporal Feature Modeling

To capture the temporal dependencies appropriately, conventional methods, such as Autoregressive Moving Average (ARIMA) [19] are widely used for the detection of static data anomalies. However, they suffer from bottlenecks and are rare used in multisensor MTS in current work. In recent years, machine learning-based methods have gained traction for anomaly detection. Local Outlier Factor (LOF) [20] detects anomalies by evaluating the distribution of time points, performing well in datasets with varying density regions. While these methods have shown excellent performance relative to statistical methods, they tend to ignore temporal dependencies and nonlinearity when dealing with high-dimensional, complex time series generated from IoT devices.

Deep learning approaches, such as AutoEncoder (AE) [21], RNNs [22], CNNs [23], [24], and Transformer are commonly employed, using prediction or reconstruction errors as anomaly indicators. AEs and its variants [25], [26] all modeled the relationship between latent variable and realized low-dimension series embedding. RNNs and LSTM are the best classic structures to model times series (sequences) because of the strong capability of integrating long-range information. Ommianomaly [3] utilized a stochastic RNN and a planar normalizing flow to generate reconstruction probabilities. But LSTM-NDT [27] relied on an LSTM-based deep neural network model to forecast data for the next timestamp. CAE-M [28] adopted attention-based bidirectional LSTM to extract the complex time features. One of notable CNN-based examples is MSCRED [29] converts MTS into spectral images and uses ConvLSTM blocks to capture the temporal patterns. Transformer has been also a viable alternative to learn the temporal features. TransAD [12] built the deep transformer network to swiftly perform inference with the knowledge of the broader temporal trends in MTS data. It introduced meta-learning to quickly optimize transformer network. Autofomer [13] reformed the transformer into a decomposition prediction structure, and decomposed and modeled the input into trend-cyclical feature and seasonal feature. It introduced the autocorrelation mechanism in place of self-attention to help model discover the similarity of subsequences based on periodical sequence. In this work, we develop the RWKV module, a novel RNN architecture that combines the strengths of RNN and transformer. The RWKV module allows for parallel and efficient sequence encoding, capturing temporal features across the previous $T - 1$ time steps [13]. This hybrid approach leverages the sequential modeling capabilities of RNNs and the parallelism and global attention mechanisms

of Transformers, making it highly effective for MTS anomaly detection.

B. Spatial Features Modeling

Unlike the single time-series sequence, MTS consist of multiple channels, each representing a value at a single time point. Early studies proposed convolutional operation to mix MTS's information. However, a single convolution kernel only integrate all signals into a predefined node feature, which neglects the relationship between signals.

After constructing the interacted graph, how to leverage these associations to update each signal's embedding is also worthy of researching. GTA [5] constructed the global graph structure and adjust the deviation of the attention mechanism. Next, the anomaly Transformer [7] renovated the self-attention mechanism to the anomaly-attention and amplified the normal-abnormal distinguishability of the association discrepancy. Currently, GNN exhibits the powerful and significant strength in message passing and aggregating neighbor features [30]. Deng et al. proposed a novel attention-based GNN approach to extract and fuse the information of a node with its neighbors based on the learned asymmetrical graph structure. GDN detected anomaly by scoring the deviation of the predicted behavior and the observed behavior at time t . Zhao et al. proposed MTAD-GAT [31], which uses graph attention network (GAT) to model the relationships between sensors. This model combines reconstruction- and forecasting-based objectives in a joint optimization framework to enhance detection accuracy. DyGraphAD [18] utilized DTW distance to build a series of dynamically evolving interseries graphs and used MixHop graph convolution to aggregate the neighbor information. MGCLAD [8] also used Gumbel-Softmax Sampling strategy to learn directed graphs and proposed a multiview graph contrastive learning method to improve representation. However, Gumbel-Softmax Sampling technique heavily relies on the temperature τ to control the discrete edge weight, which can make it difficult to enforce graph sparsity effectively.

III. METHODOLOGY

The proposed STWGR framework is designed to address the challenges of multivariate time-series anomaly detection in IoT systems by balancing three key objectives: 1) high detection accuracy; 2) low latency; and 3) reliable results. The framework integrates two core modules D-RWKV and TaGAA to efficiently model spatio-temporal dependencies and dynamically infer relationships between sensor signals. The Fig. 2 is illustration of the architecture overview of STWGR.

A. Problem Statement

Given a MTS $S \in \mathbb{R}^{N \times T}$, collected over T times ticks with N sensors. Note that N sensors could be viewed as N channels (nodes) in the method modeling for convenience. The test set is collected over a different span of time ticks. We further define $\bar{x}_t$ as the feature vector of the t timestamp for each sensors. It is filled with previous window values: $\bar{x}_t := [e_{i, \cdot}, i \in \{t - M + 1, t - M + 2, t\}]$, where $e_{i, n}$ is the normalized

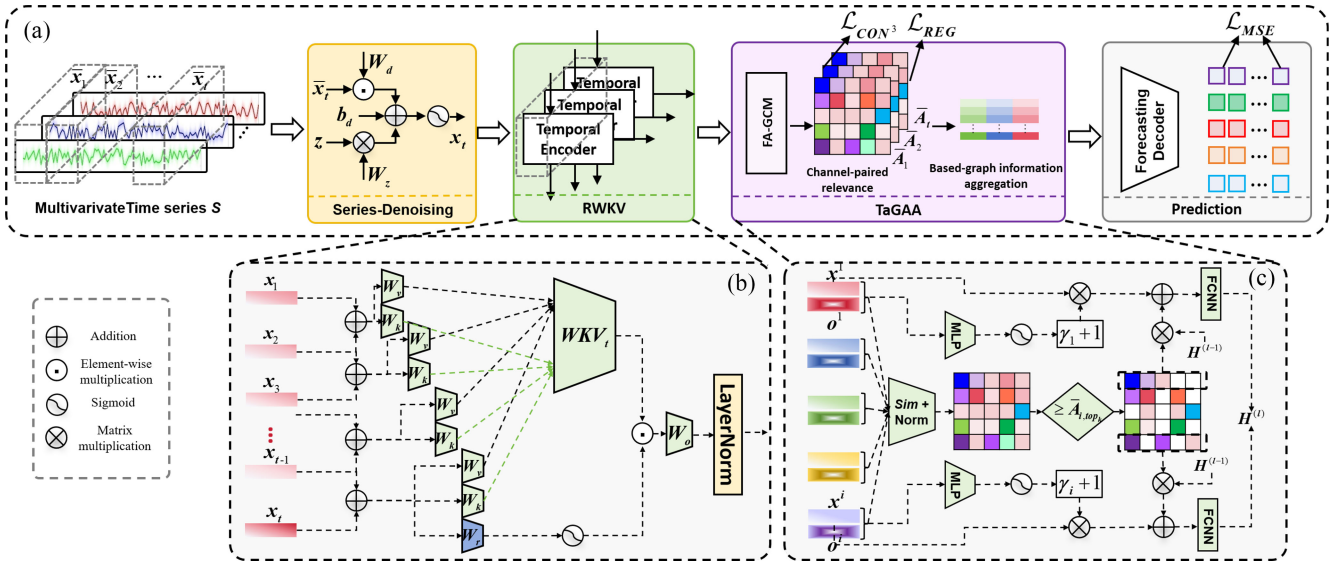


Fig. 2. Overview of our proposed framework, (a) STWGRU. It consists of four components. The Series-Denoising Receptance Weighted Key Value (D-RWKV) preliminarily filters noisy signals and quickly learns historical features via RWKV (b). The Targeted-awareness Graph Adaptive Aggregation (c) constructs a robust dynamic graph for each sequence and graph convolutional layers with adaptive feature retention. The Prediction Decoder can forecast the next time series values.

value of n sensor in the t timestamp, M is the filled window size. Moreover, when $i \leq 0$, the $e_{i,n}$ is filled with 0. MTS anomaly detection is defined as a problem that determines whether an observation from a certain task at a certain time is anomalous or not.

B. Series-Denoising Receptance Weighted Key Value Module

For the anomaly detection task, it is crucial to capture and learn vital information from historical sequences. However, traditional methods face challenges, such as the vanishing gradient problem in RNNs and the high computational complexity of Transformers. Inspired by [32], we construct a series-denoising receptance weighted key value module (D-RWKV), which achieves both sequence denoising and historical feature learning simultaneously. The D-RWKV first employs a simple yet effective function to filter the noisy disturbance

$$x_t = \bar{x}_t - \text{Sigmoid}(W_d \cdot \bar{x}_t + W_z \odot z + b_d) \quad (1)$$

where x_t denotes the MTS sequence vector after filtering the noise. The $\{W_d, W_z\} \in \mathbb{R}^{M \times M}$, $b_d \in \mathbb{R}^M$ are learnable parameters. The vector $z \in \mathbb{R}^M$ are sampled from the Gaussian distribution $\mathcal{N}(0, \sqrt{\delta_t}I)$ to simulate the white noise. $\sqrt{\delta_t}$ is inferred based on the input sequence. We further model the time-dependence correlations and generate the historical representation,

$$r_t = W_r \cdot (\mu_r x_t + (1 - \mu_r)x_{t-1}) \quad (2)$$

$$k_t = W_k \cdot (\mu_k x_t + (1 - \mu_k)x_{t-1}) \quad (3)$$

$$v_t = W_v \cdot (\mu_v x_t + (1 - \mu_v)x_{t-1}) \quad (4)$$

where $r_t \in \mathbb{R}^{1 \times M}$ denotes the receptance vector acting as the acceptance of past information. And $k_t \in \mathbb{R}^{1 \times M}$ denotes Key vector analogous to K in traditional attention mode $\text{Attn}(Q, K, V)$. $v_t \in \mathbb{R}^{1 \times M}$ is also a vector analogous to V in the same attention one. Moreover, $\{W_r, W_k, W_v\} \in$

$\mathbb{R}^{M \times M}$ denote the three dependent project parameter matrices. $\{\mu_r, \mu_k, \mu_v\} \in \mathbb{R}^{M \times M}$ are three trainable variable. They are used in a linear combination of x_t and x_{t-1} , to achieve a simple time mixing. Based on these vectors, D-RWKV module designed a linear complexity attention mechanism:

$$wkv_t = \frac{\sum_{i=1}^{t-1} e^{-(t-1-i)w+k_i} v_i + e^{u+k_t} v_t}{\sum_{i=1}^{t-1} e^{-(t-1-i)w+k_i} + e^{u+k_t}} \quad (5)$$

where the WKV computation, wkv_t , plays the role of $\text{Attn}(Q, K, V)$ [32] in vanilla Transformers without incurring a quadratic cost (i.e., QK^T product operation) as feature's interactions are between scalars. The $e^{-(t-1-i)w+k_i} = e^{-(t-1-i)w} \cdot e^{k_i}$ is a straightforward exponential-additive combination of time decay and key attention. The w , also known as "time decay," is one trainable parameters with the initialized setting of $-5 + 8 \cdot (i/[d-1])^{0.7 + (1.3m/[M-1])}$, where d is the embedded dimension. Intuitively, it is the learnable discount factor applied to previous tokens over time. Besides, the u , also known as "bonus," is set to $0.5(((m+1) \bmod 3) - 1) + \log 0.3$ [33]. It is the special weighting applied to the current token in (5). This helps the model pay more attention to the current token and circumvents any potential degradation of w . The alternating zigzag pattern initially creates subtle variations in the tensor elements, which are intend to help the model treat different timestamps of the embedding distinctively. Next, for the target position t , we utilize the r_t and wkv_t to compute the memory-state embedding of the n sensor in the l block as follows:

$$o_t = \text{LN}(W_o \cdot (\text{Sigmoid}(r_t) \odot wkv_t)) \quad (6)$$

where $\text{Sigmoid}(r_t)$ makes the receptance vector nonlinearly project the range $[0, 1]$. Besides, interactions are multiplicative inside a given time-step and summed over different time-steps of the positional interval of $[1, t]$. The $\text{LN}(W_o(\cdot))$ performs the projection and LayerNorm operations. As t increases, o_t is

dependent on a long history, represented by summation of an increasing number of terms.

C. Targeted-Awareness Graph Adaptive Aggregation

The proposed framework effectively captures temporal dependencies in MTS sequences through the D-RWKV module, which progressively accumulates information from the previous $t - 1$ time steps using adaptive attenuation weights and constructs the historical representation o_t for each channel. However, in complex IOT systems, there is functional coupling causing that different monitoring sensors also produce spatial correlation. Learning the underlying correlations can help detect system anomalies in advance, improving the efficiency and accuracy of anomaly detection. Thus, we propose the Targeted-awareness graph adaptive aggregation (TaGAA) module, including the graph structure reasoning and based-graph information aggregation. According to the original and historical representations, we leverage the graph structure learning strategy to reason the relations among the sensors. To construct the relation graph in the t time, the first operation is to calculate channel-paired relevance, as following:

$$\bar{A}_{ij} = \frac{\text{Sim}([o_t^i, x_t^i], [o_t^j, x_t^j])}{\sum_{k \in \mathcal{N}^{\setminus \{i\}}} \text{Sim}([o_t^i, x_t^i], [o_t^k, x_t^k])} \quad (7)$$

where Sim presents a metric function. The cosine similarity is adopted as the Sim . To obtain relations between the historical feature and the current sequence feature, we adopt a concatenation operation to combine these representations, $[o_t^i, x_t^i] \in \mathbb{R}^{w+d}$. In addition, $\bar{A}_{ij}$ denotes the correlation value between the sensor i and the sensor j in t times. But employing the Sim operation to calculate weights is all than 0, $\bar{A}_{ij} > 0 \forall \{i, j\} \in \{1, 2, \dots, N\}$. This denotes that $\bar{A}_{ij}$ has the deviation or error for unrelated sensor pairs, which contaminates the sensor's representation. On the other hand, the symmetric correlations are invalid, i.e., $\bar{A}_{ij} \neq \bar{A}_{ji}, \exists \{i, j\} \in \{1, 2, \dots, N\}$. We further conduct targeted perception

$$A_{ij} = \begin{cases} \bar{A}_{ij}, & \text{if } \bar{A}_{ij} \geq \bar{A}_{i, \text{top}-k} \\ 0, & \text{otherwise} \end{cases} \quad (8)$$

where $\bar{A}_{i, \text{top}-k}$ denotes the $\text{top} - k$ biggest value in the i th row weight vector $\bar{\mathbf{A}}_i$. When $\bar{A}_{ij} \geq \bar{A}_{i, \text{top}-k}$, the weight A_{ij} is equal to $\bar{A}_{ij}$, otherwise 0. By utilizing 8, the most relevant weights are reserved to construct the adjacency matrix $\mathbf{A}$ of targeted-awareness graph $\mathcal{G}$ of the t th timestamp.

The graph $\mathcal{G}$ can explicitly characterizes the correlations between sensors (node for graph learning) with the above pipeline operation. Next, we need to model and represent the influence of these correlations to node's embeddings, and further explore how associated nodes change the subsequent behaviors of certain node. Most of GCN [34], [35], e.g., graph isomorphism network (GIN) [36], GAT [37], and ChebyNet [38] have demonstrated exceptional capabilities in information aggregation and node representation learning. The main architecture of GCN includes the two components, as following:

$$h_v^{(l)} = \text{COMBINE}^{(l)}(h_v^{(l-1)}, a_v^{(l)}) \quad (9)$$

$$a_v^{(l)} = \text{AGGREGATE}^{(l-1)}(\{h_u^{(l-1)}; u \in \mathcal{N}(v)\}) \quad (10)$$

where $\text{AGGREGATE}(\cdot)$ presents integration and processing of neighbor features. $\text{AGGREGATE}(\cdot)$ presents to embed whole neighbor features into meta node's representation. Hence, we design the graph adaptive aggregation module based on GCN architecture for MTS sequence

$$h_i^{(l)} = f_{\text{proj}} \left((1 + \gamma_i) h_i^{(l-1)} + \sum_{j \in \mathcal{N}(i)} A_{ij} h_j^{(l-1)} \right) \quad (11)$$

$$\gamma_i = \text{Sigmoid}(\text{MLP}_1([o^i, x^i])) \quad (12)$$

where $h_i^{(l)}$ is the feature vector of node i at the l th layer. We set $h_i^l = x^i \in \mathcal{X}$ for the i th time-step embedding. A_{ji} represents prior relevance that the j neighbor imposes on i node. And $\mathcal{N}(i)$ is the neighbor set of the i th node. $f_{\text{proj}}(\cdot)$ represents the injective multiset function composed of a single full-connected network. We let $\gamma_i \in \Gamma = \{\gamma_1, \gamma_2, \dots, \gamma_N\} \in \mathbb{R}^N$ is harmonic scalar to retain the previous feature of i th node, where the enhancement scalars of node's own features are computed by the MLP layer with Sigmoid activation. Compared with the vanilla GCN methods [36], we not only transfer the invariant to the learnable value to generate the flexible node's representation, aggregate the neighbor features via corresponding related weights.

D. Model Decoding and Optimization

1) *Time Series Forecasting Decoder*: For the anomaly detection in MTS, it is a general method that forecasts channel results at the $t+1$ timestamp based on the embedded features at the t timestamp [12], compared to the classical reconstruction input X_t method. In this work, we adopt the forecasting next-value method to construct the MTS Decoder, as following:

$$Y = \text{ReLU}(\text{MLP}_2(\text{ReLU}(\text{BN}([H, O]))) \quad (13)$$

where we concatenate the D-RWKV module output O and the TaGAA module one H as the final embedding, which aims to aggregate the historical and interacted information. BN denotes the BatchNorm operation to increase the uniformity of stability for the cascaded embeddings $[H, O] \in \mathbb{R}^{2d}$. The ReLU activated function and MLP network are used together to predicted results $Y \in \mathbb{R}^N$.

2) *Context Consistency Constraint*: We have modeled dynamic graph to explicitly express the directed relation of node o^i to o^j for each time sequence. Since node embeddings vary across timestamps, the relationships between node pairs may also differ over time. However, in an IoT monitor system, most directed relationships should be stable during normal states according to prior experience and observation. Some relations could appear to change when the system is in a transitional state between normal and abnormal, or an abnormal state. Thus, the relation graphs computed from sliding windows of time sequences may fail to accurately reflect the true relationships between nodes. To ensure the relation consistency and help our framework learn the actual

underlying relations, we introduce the context consistency constraint (CON^3) to amend these relations of timestamps:

$$\mathcal{L}_{\text{CON}^3} = \frac{1}{T} \sum_{t=1}^T \left(1 - \|Y^t - Y^{t-1}\|_1\right) \cdot \|\mathcal{G}^t - \mathcal{G}^{t-1}\|_1 \quad (14)$$

where Y^t and $\mathcal{G}^t = A^t$ are the groundtruth and directed relation graph over the t time. The $\mathcal{L}_{\text{CON}^3}$ aims to minimize the gaps of contiguous graphs when their corresponding Y are same or tiny difference. In other words, the error value of $\|\mathcal{G}^t - \mathcal{G}^{t-1}\|_1$ is involved in the loss as the item $1 - \|Y^t - Y^{t-1}\|_1 > 0$. Using $\mathcal{L}_{\text{CON}^3}$ can deftly realize the simultaneous optimization of graph consistency and difference.

3) *Whole Loss Function*: Though most of the approaches commonly adopt the autoregressive prediction, our work predicts a one-step ahead prediction based on the $t - 1$ time representation. This is because RWKV historical encoder would occur the information disclosure from the future if employing the autoregressive prediction, which is noncausal. Finally, the whole loss function is as following:

$$\mathcal{L} = \mathcal{L}_{\text{MSE}} + \alpha \mathcal{L}_{\text{CON}^3} + \beta \mathcal{L}_{\text{REG}} \quad (15)$$

where the prediction loss $\mathcal{L}_{\text{MSE}}$ denotes the Mean Square Error (MSE) to minimize the error between the forecasting value and the ground truth. In addition, $\mathcal{L}_{\text{REG}} = (1/T) \sum_{t=1}^T |A^t|$ is a regularization constraint of graphs via the l_1 norm to further reinforce the sparsity of related graphs. We introduce the hyperparameters α and β to adjust the utility of graph losses and achieve the best results for different tasks.

IV. IMPLEMENTATION DETAILS

To investigate the efficiency and reliability of the proposed method in diversified complex IoT systems, we adopt six publicly available benchmarks for MTS anomaly detection. we introduce their characteristic and show the used the size of train and test of each dataset in supplemental materials. We compare the performance of our proposed framework with fourteen popular anomaly detection methods.

We perform data standardization and divide all data sets into train and test components by referring to the previous work [12]. The models are trained using the AdamW optimizer with different learning ratio and the same step scheduler with a step size of 0.5 for each dataset. To realize the best performance for MTS anomaly detection, we conducted the optimized searching for learning ratio, epochs, top-K, which are reported in Table I. The two hyperparameters α , β of $\mathcal{L}_{\text{CON}^3}$ and $\mathcal{L}_{\text{REG}}$ will be discussed in section V-C (1). We gave the optimal the weights in Table I. Different to other baselines, we do not project the sequence into a high-dimension latent space and keep the dimension of key embeddings (o_t and h_t) are equal to window size M . Thus the stable window size are searched and recorded for six datasets. The the mean of $5 \times$ experiments using different seeds is reported.

V. EXPERIMENTAL RESULTS AND ANALYSES

In this section, we conduct comprehensive experiments to mainly answer the following questions.

TABLE I
CORE HYPERPARAMETERS IN SIX DATASETS

Dataset	M	Learning Ratio	Epochs	Top-K	α	β
SMAP	10	0.001	4	15	0.005	0.001
SWaT	10	0.001	7	30	0.001	0.0001
SMD	32	0.0001	10	30	0.005	0.001
MSDS	20	0.00001	10	4	0.01	0.0001
MSL	10	0.002	5	30	0.01	0.001
PSM	32	0.0001	4	15	0.005	0.001

- How does our model perform compared with baselines?
- How do the key modules contribute to our model?
- How are some important hyperparameters selected?
- Is our model efficient?
- Can learned graphs represent the MTS dependencies?
- How well does STWGRL predict the sequence?

A. Performances Comparison

We conduct comprehensive experiments in six datasets to compare with several SOTA baselines. These baselines' scores are extracted from previous works [7], [8], [12]. All experimental results are recorded in Tables II and III. Our method realize comprehensive and significant improvements in terms of F1 metric, such as the highest improvement (3.87%) on MSDS. STWGRL achieved the best F1 score (98.49%) on SMAP dataset, obtaining the solid gain of 1.86% than the Anomaly Transformer's result (96.69%). This suggests that STWGRL can detect various anomalies in IoT systems with high accuracy and strong generalization ability. Evident improvements should owe to the long-term dependent modeling and the interaction learning between different signals.

Besides, some extra insights are summarized as:

1) For precision (P) and recall (R) metrics, our model achieved the competitive performances compared with several SOTA baselines, such as 98.86% on SMAP, 99.72% (best: 100%) on MSDS. This presents that false anomalies need to further analysis to reduce the false alarm rate (1-P). However, we can find that most of baselines have a diametrically opposed accuracy and recall. MSCRED obtained the 99.92% precision but only 67.70% on SWaT. CAE-M obtained the 99.98% recall but only 77.51% precision. Our method strikes a stable tradeoff between two metrics. Better recall suggests that STWGRL is proficient in detecting infrequent anomalies within time series.

2) For GNN-based methods, MTAD-GAT and GDN only learn a single signals' relationships based on the global time series. It is difficult to response the flexible changes, which gets the lower F1 scores (83.42% on SMD, 66.22% on PSM). DyGraphAD enhances its performance by capturing the relevances among multiple series within a narrow window range. Nevertheless, our method consistently surpasses DyGraphAD in terms of performance. This underscores a significant advantage of our approach: explicitly modeling

TABLE II

PERFORMANCE COMPARISON OF ACGSL WITH BASELINES ON SMAP, SWAT, AND SMD. THE BEST AND SECOND-BEST SCORES ARE HIGHLIGHTED IN BOLD AND WITH UNDERLINES, RESPECTIVELY

Methods	SMAP			SWaT			SMD		
	P	R	F1	P	R	F1	P	R	F1
LSTM-NDT [27]	85.23	73.26	78.79	77.78	51.09	61.67	97.36	84.40	90.42
DAGMM [26]	80.69	98.91	88.88	99.33	68.79	81.28	91.03	99.14	94.91
OmniAnomaly [3]	81.30	94.19	87.28	97.82	69.57	81.31	88.81	99.85	94.01
MSCRED [29]	81.75	92.16	86.64	99.92	67.70	80.72	72.76	99.74	84.14
MAD-GAN [39]	81.57	92.15	86.54	95.93	69.57	80.65	99.91	84.40	91.50
MTAD-GAT [31]	79.91	99.91	88.80	97.18	69.57	81.09	82.10	92.15	86.83
CAE-M [28]	81.83	85.67	88.27	96.96	69.56	81.01	90.82	96.71	93.67
GDN [16]	74.80	98.91	85.18	96.96	69.56	81.01	71.70	99.74	83.42
GTA [5]	89.11	91.76	90.41	94.83	88.10	91.34	85.86	88.72	87.27
DyGraphAD [18]	93.39	96.55	94.94	94.29	90.42	92.31	95.14	98.19	<u>96.55</u>
Anomaly Transformer [7]	94.13	99.40	<u>96.69</u>	91.55	96.73	<u>94.07</u>	89.40	95.45	92.33
Autoformer [13]	90.40	58.62	71.12	89.85	95.81	92.74	88.06	82.35	85.11
MGCLAD [8]	93.42	92.71	93.07	94.81	89.11	91.87	95.17	93.52	94.34
TranAD [12]	80.43	99.99	89.15	97.60	69.97	81.51	92.62	99.74	96.05
Ours	98.86	98.14	98.49	95.79	97.97	96.87	95.99	99.74	97.83
Improvement			↑1.86%			↑2.98%			↑1.33%

TABLE III

PERFORMANCE COMPARISON OF ACGSL WITH BASELINES ON MSDS, MSL, AND PSM. THE BEST AND SECOND-BEST SCORES ARE HIGHLIGHTED IN BOLD AND WITH UNDERLINES, RESPECTIVELY

Methods	MSDS			MSL			PSM		
	P	R	F1	P	R	F1	P	R	F1
LSTM-NDT [27]	99.99	80.12	88.96	62.88	99.91	77.21	73.62	89.92	80.96
DAGMM [26]	98.91	80.26	88.61	73.63	99.97	84.82	76.09	92.45	83.48
OmniAnomaly [3]	100.00	79.64	88.67	78.48	99.24	87.65	91.61	71.36	80.23
MSCRED [29]	100.00	79.83	88.78	89.12	98.62	93.63	-	-	-
MAD-GAN [39]	99.82	61.07	75.78	85.16	99.30	91.69	88.63	96.51	92.40
MTAD-GAT [31]	99.19	79.64	88.35	79.17	98.24	87.68	77.67	89.99	83.38
CAE-M [28]	99.08	84.39	91.15	77.51	99.98	87.33	-	-	-
GDN [16]	99.89	80.26	89.00	93.08	98.92	95.91	56.72	79.54	66.22
GTA [5]	-	-	-	91.04	91.17	91.11	90.57	92.88	91.72
DyGraphAD [18]	-	-	-	97.01	94.86	<u>95.92</u>	-	-	-
Anomaly Transformer [7]	-	-	-	92.09	95.15	93.59	68.35	94.72	79.40
Autoformer [13]	-	-	-	77.27	80.92	79.05	99.08	88.15	93.29
MGCLAD [8]	-	-	-	95.10	95.89	95.49	96.17	91.39	<u>93.72</u>
TranAD [12]	99.99	86.26	<u>92.62</u>	93.08	99.99	94.94	67.81	90.30	77.45
Ours	99.72	90.13	94.68	99.93	93.69	96.71	95.52	95.59	95.56
Improvement			↑3.87%			↑0.82%			↑1.96%

dynamically evolving inter-relationships without being constrained by a predefined window.

B. Ablation Studies

1) *Module Ablation Analysis*: To verify the importance of each module, we compared it with its following derivatives.

- 1) *w/o Denoising*: To investigate the necessity of filtering specific noise data in the sequence, we straightly remove the denoised layer from D-RWKV module.
- 2) *w/o D-RWKV*: To verify the effect of historical representation in the process of graph learning and MTS prediction, we remove the D-RWKV module and set $H = X$ in (13).

- 3) *w/ LSTM*: LSTM is the classical structure to learn the context information and structure hidden features. To compare LSTM with D-RWKV module, we replace the D-RWKV with a single LSTM layer.
- 4) *w/o TaGAA*: To check the necessity of modeling signals' spatial feature interaction, we remove the TaGAA module and let the aggregated representation $O = X$ in (13).
- 5) *w/o L_{CON^3}* : To further investigate the effect of the context consistency constraint in graph construction process, we keep the change of neighbor interacted graph free by setting α as 0 in (15).
- 6) *w/o L_{REG}* : To evaluate the effect of graph sparsity via l_1 norm operation for refining the core edges, we give up using the loss L_{REG} by setting β as 0 in (15).

Table IV shows the ablation results of STWGRL on six datasets. From this table, we gained the following observations.

- 1) The w/o Denoising and w/o D-RWKV variants are inferior to the whole framework, verifying the importance of filtering the noise features and modeling the temporal information.
- 2) Using LSTM cell as the extractor of historical features is more restricted than D-RWKV module. The average performance of w/ LSTM (92.68%) decreases by 4.15% than the w/ D-RWKV one (96.69%), which indicates that mixing temporal features via decayed attention is reliable.
- 3) STWGRL surpassed w/o TaGAA variant, indicating that the designed signals' relations modeling and graph reasoning can indeed augment the signal feature and improve the model predicted capacity.
- 4) w/o L_{CON^3} is inferior to our framework, which exhibits the validity of keeping the consistency between neighbor graphs. L_{REG} is also necessary for most of datasets except MSDS dataset. The F1 score of w/o L_{REG} outperforms our framework, 95.92% versus 94.68%. This can be causing by the lesser top-K setup in MSDS.

2) *Comparative Analysis for GNN*: It is crucial for MTS's prediction to learn the interacted representation based on directed graph. In this work, we proposed targeted-aware feature aggregation strategy to incorporate node's features. To evaluate the efficiency variance between our strategy and some classic GNN methods, such as GCN [34], GAT [37], GraphSAGE [35], ChebyNet [38], and GIN [36], we conducted comprehensive experiments on six dataset and reported results in Table V. The results present that the proposed strategy is superior to most of GNN methods on different datasets. For instance, the performance improves by up to 24.35% (76.14% via GAT versus 94.68% via ours) on SMAP dataset. Besides, through adaptively learning mixed parameter γ_i for each node, the original feature is preserved with more broad weights ($1+\gamma_i \in (0, 2)$). Thus, the performance (avg.F1: 94.63%) expresses better than GIN method (avg.F1: 96.69%) via an invariant γ for all node.

C. Parameter Analyses

1) *Influence of L_{CON^3}* : The context consistency constraint (L_{CON^3}) aim to restrain the difference of contiguous graphs based on the signal's values. Stronger constraint could cause the oversmooth of contiguous graphs and lose the diversities of interaction. Thus, to investigate the influence of L_{CON^3} , we set its hyperparameter α with different weight, i.e., $\alpha \in \{0, 1e-5, 5e-5, 0.0001, 0.0005, 0.001, 0.005, 0.01, 0.05, 0.1, 0.5, 1\}$. we exhibit the performance comparison with different metrics on all datasets, respectively, shown in Fig. 3(a)–(c). These figures distinctly show that different dataset have different constraint strength requirements. PSM acquired the best F1 score when $\alpha = 0.005$, but SWaT and SMD required α as 0.001 and 0.5, respectively. In addition, for MSDS and SMAP dataset, they are insensitive to the α weight, keep the F1 performance with stable range [94.11%, 94.68%] and

TABLE IV
ABLATION STUDIES OF THE ACGSL IN TERMS OF AUC AND F1 SCORES

Methods	F1 measures (%)					
	SMAP	SWaT	SMD	MSDS	MSL	PSM
w/o Denoising	97.56	92.55	97.55	95.90	95.47	94.76
w/o D-RWKV	97.13	93.37	97.39	76.18	93.67	93.18
w/ LSTM	96.64	90.51	94.89	89.67	92.22	92.17
w/o TaGAA	96.34	93.75	97.42	90.17	93.68	94.52
w/o L_{CON^3}	96.72	92.94	95.46	94.11	94.40	89.17
w/o L_{REG}	86.18	94.92	97.18	95.92	95.87	93.04
Ours	98.49	96.87	97.83	94.68	96.71	95.56

TABLE V
PERFORMANCE COMPARISON OF DIFFERENT GNN AGGREGATORS

Methods	F1 measures (%)					
	SMAP	SWaT	SMD	MSDS	MSL	PSM
GCN [34]	82.70	89.88	86.32	88.95	89.36	89.94
GAT [37]	80.95	93.29	84.39	76.14	85.08	82.00
GraphSAGE [35]	92.12	94.37	91.21	77.10	90.46	79.83
GIN [36]	96.72	92.94	95.46	94.11	94.40	90.38
Ours	98.49	96.87	97.83	94.68	96.71	95.56

[97.32%, 98.49%], respectively. In contrast, the performance of PSM fluctuated significantly from 89.17% to 95.56%. The larger the weight ($\alpha > 0.005$), our model's ability to identify multiple anomalies in the PSM dataset continues to decline (the precision increases, but the recall decreases).

2) *Sensitivity of Top-K*: The adoption of the top-K strategy effectively facilitates the removal of noise and unnecessary interactions, while simultaneously capturing potential cause-effect relationships among sensors. We design experiments herein to examine the impact of various top-K configurations on model performance. Given the variability in the number of sensors across datasets, the percentage of retained edges is used to standardize the top-K selection process. The range of percentages spans from 20% to 100%, inclusive, in increments of 20%. As illustrated in Fig. 4, each dataset corresponds to an unique percentage of edges (top-K). For example, achieving a high F1 score on the MSDS dataset requires retaining 20% of edges per sensor. In contrast, the SMD dataset performs best when 80% of interactions are retained (top-K=32).

D. Model Complexity

To evaluate the efficiency of the proposed model, we calculate and compare the number of FLOPs and model parameters of STWGRL with different window sizes (M) to other baselines. Note that we fixed the same input size for baselines with window sizes (M=10) and exhibit the results in Table VI. It can be observed that STWGRL (M=10) has fewer parameters and FLOPs compared to most baselines. The number of parameters is reduced by more than 10 $\times$ compared to the second fewest model (DAGMM).

For attention-based methods, while Anomaly Transformer utilizes attention operation to model temporal patterns by separately extracting prior and series association, this greatly increases its complexity, $\mathcal{O}(M^2d)$ for a transformer block.

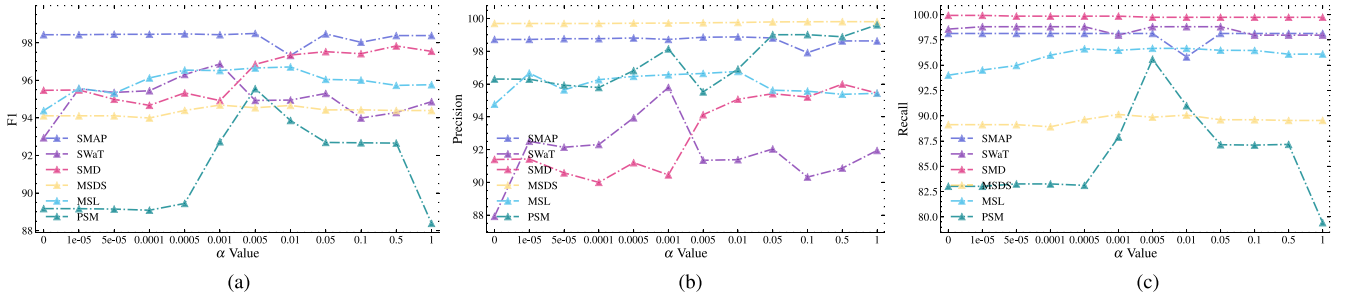


Fig. 3. Empirical analysis for hyperparameter α of L_{CON3} on six datasets. (a) α F1. (b) α Precision. (c) α Recall.

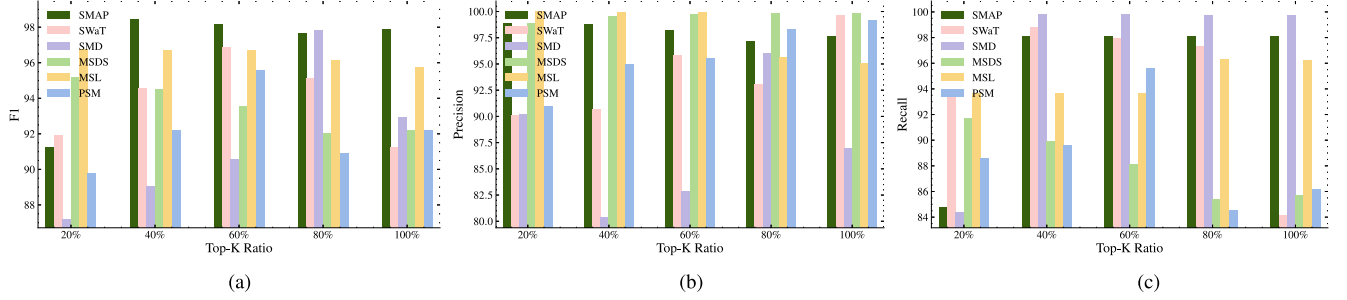


Fig. 4. Empirical analysis for the selection of top-K on several datasets. (a) Top-K F1. (b) Top-K Precision. (c) Top-K Recall.

TABLE VI
FLOPS AND PARAMETER QUANTITIES OF STWGR AND BASELINES. THE FEWEST VALUES ARE IN BOLD AND THE SECOND FEWEST ARE UNDERLINED. GMM: GAUSSIAN MIXTURE MODEL

Methods	FLOPs (M)	Param(K)	Key Coempts
DAGMM	5.20	87.51	MLP, GMM
OmniAnomaly	85.12	2575.20	GRU
MTAD-GAT	5.69	4250.25	Graph, Joint criteria
GTA	200.32	2963.81	Attention, Graph, MLP
Anomaly Trans	143.99	2403.12	Attention, Joint criteria
MGCLAD	<u>1.61</u>	92.58	Graph, Attention
Ours(M=10)	1.52	8.12	
Ours(M=20)	3.06	<u>17.09</u>	RWKV, Graph
MM Ours(M=30)	4.63	27.66	

However, RWKV employs a more efficient linear scaling, $\mathcal{O}(Md)$, and alleviates the quadratic scaling in Transformer. Moreover, following [22], [33], the FLOPs of D-RWKV can be approximated as $6(MD + 13D^2L)$, where L represents the number of layers. It is noteworthy that RWKV's FLOPs are independent of the context length, unlike regular transformers. The FLOPs of TaGAA module is $2(2MD) + 2D$. Compared to GRU and graph-based methods, the TaGAA module carefully considers all node-pair relations while using only a single graph convolution layer for feature integration. This design makes it simpler and more efficient.

E. Context Graph Analysis

1) *Learned Graph Visualization*: We adopt the MSDS dataset as the case study to concretely explore the relationships and differences among signals. Fig. 5 gives the real signals' values (the bottom row figure) sampled from MSDS test set with length $T = 2000$ and the correlated matrix maps (the

top row figure) at 4 (2 normal and 2 anomaly) random timestamps. The color is more red presents the weight is more large, otherwise smaller. Comparing the graph maps between normal and anomaly states, the maps are distinctly different. Taking the first signal X_1 as an example, X_1 have high weights with the first five signals, $A_{1,1:4}^2 > 0.2$ as the normal state. On the contrary, the high related weights transfer these signals 4:7 as anomaly state. This presents that our model can capture dynamic changes for signal's relevance when IOT state happens abnormal. However, it is easy to observed that the maps also changes at different times of the same state, such as $A_{1,4}^1 < A_{1,4}^2$. The can be a normal fluctuation as the signal fluctuates, which is verified by further experiments.

2) *Graph Consistency Visualization*: To explore the above phenomenon and investigate whether neighbor graphs exist consistency, we calculate the different degree of neighbor time for each signal: 1) compute the mean of each signal's correlated weights, i.e., $A_i^t = (1/M) \sum_{j=1}^M A_{i,j}$ and 2) compute the different degree as the neighbor time, i.e., $\Delta_i^{t-1 \rightarrow t} = |A_i^{t-1} - A_i^t|$. We perform these calculations on MSDS and PSM test sets and exhibit the results in Fig. 6. Note that the anomaly timestamps are labeled in the bottom row with the column label *Lab*. We can see that the difference of signal correlation changes frequent as the continual changed states for MSDS. It is difficult to model long-range consistency for model, which can explain the phenomenon of different graphs of the same state in the Fig. 5. However, for PSM dataset, the state is the same, the $\Delta_i^{t-1 \rightarrow t}$ is close to zero, i.e., the color is more blue. This presents the relationships of each signal exist consistency. STWGR can model the relationships and keep the relations permanent. Relationships change only when something is about to go wrong.

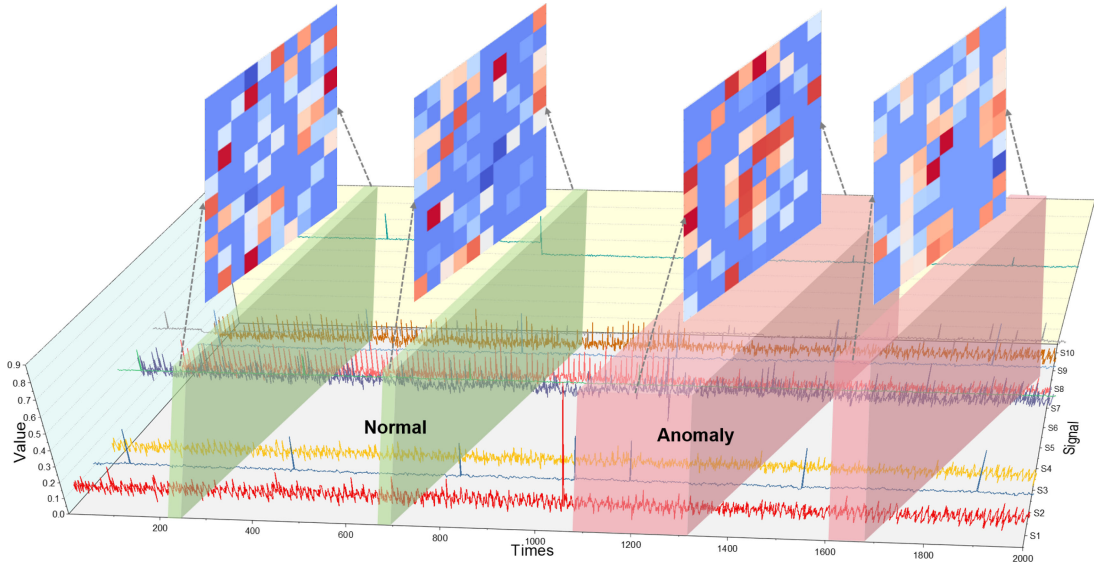


Fig. 5. Empirical analysis for learned graphs on the MSDS test set. The top row gives four heat maps to present relations among nodes at normal and anomaly timestamps. The bottom row shows the values of different signals over a certain time period, as well as the interval of abnormal moments.

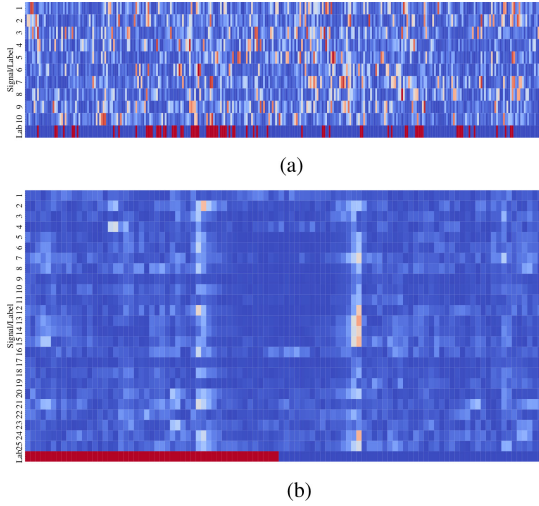


Fig. 6. Edge-weight difference maps of neighbor time point for each signal in MSDS and PSM test set. (a) MSDS. (b) PSM.

F. Case Studies

We compared the predicted and observed signal values in Fig. 7 to intuitively investigate performances of state prediction and anomaly detection of STWGRL in term of multivariate time-series. The authentic and predicted values are marked via black and red color, respectively. Blue line and regions denote the time range occurring anomaly. The named “Anomaly Score” subfigures give the computed anomaly threshold. STWGRL can fit the signals’ values well, in special the expected spikes in the signal 0 and 1. This presents that the differentiated constraint $\mathcal{L}_{\text{CON}^3}$ can help our model construct the lived graph and change the output. Accurate predictions are beneficial to timely finding anomalies and further analyzing the reason behind the phenomena. However, the random oscillations are existed in predicted values of different signals, lead to the bad fitness. Moreover, the ability

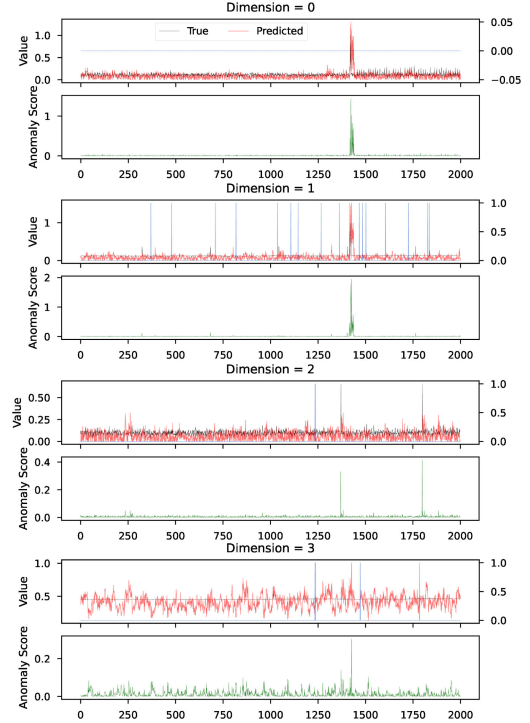


Fig. 7. Predicted and ground truth labels for the MSDS test set, where each dimension denotes one signal.

to detect accurately each anomaly is inadequate according to the “Anomaly Score” subfigures. The drawback seems to be concealed by the point-adjust strategy.

VI. CONCLUSION

This article proposes a novel spatio-temporal weighted graph reason learning framework, termed STWGRL, for MTS anomaly detection in IoT devices. The framework leverages

TaGAA module to reason the relationships between signals and integrate the message passing, constructing robust spatio-temporal representation. The series-denoising receptance weighted key value module (D-RWKV) mixes the prior feature with the linear attention. Two constraints further strengthen the discrimination between normal and abnormal graphs. Experiments on six datasets have shown the effectiveness of the proposed method in anomaly detection tasks.

Limitations and Future work: Although STWGRL achieves satisfactory results in various real-world datasets, it has some tradeoffs between the predicted details and the detection performance. Random oscillations are still observed in some predicted signals, which can hinder the precise calculation of anomaly thresholds. The next improvement should pay more attention to discuss the correlation between the learned graphs and actual system behaviors. Moreover, consistency and stability of the learned graphs within a time segment require further reinforcement. We also plan to embed our model into real-time anomaly detection systems via hardware–software co-design and online incremental learning mechanisms to verify the generalizability to IoT applications.

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